PharmAchieve

EYE HEALTH

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

AMD	Age-Related Macular Degeneration
AREDS	Age-Related Eye Disease Study
HSV	Herpes Simplex Virus
HZ	Herpes Zoster
MGD	Meibomian Gland Dysfunction

OBJECTIVES

For each eye condition (red eye, eyelid conditions and age-related macular degeneration):

1. Explain the pathophysiology
2. Identify the clinical presentation, diagnostic methods, and red flags
3. Understand goals of therapy
4. Understand management with non-pharmacological and pharmacological measures
5. Identify monitoring parameters

RED EYE

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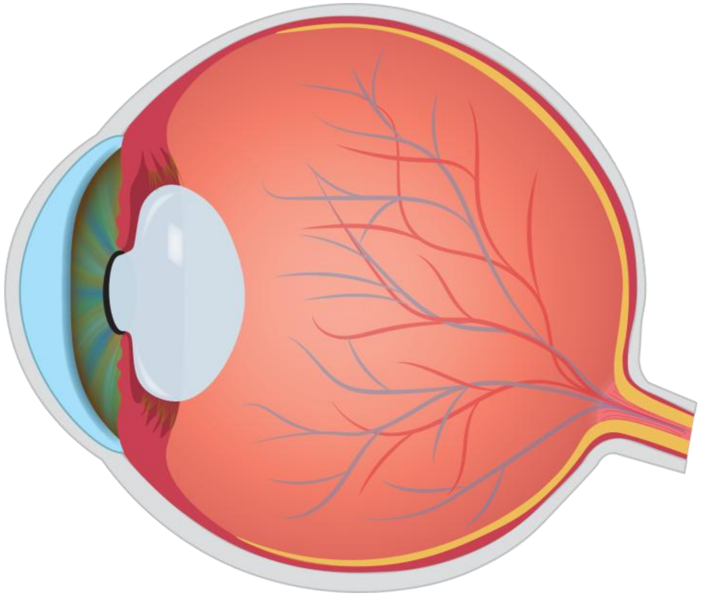
PATHOPHYSIOLOGY

- Red eye indicates inflammation resulting from dilated surface eye blood vessels
- Conjunctivitis is the most common cause of red eye
- There are many possible causes of red eye:
  - Infections (viral or bacterial): conjunctivitis, keratitis, blepharitis
  - Allergies
  - Ocular inflammation: iritis, uveitis scleritis
  - Glaucoma
  - Dry eyes: vitamin A deficiency, Sjogren's syndrome
  - Eyelid conditions: blepharitis, stye (hordeolum)
  - Irritants: ophthalmic drugs, smoke
  - Injuries: corneal abrasions, foreign bodies
  - Ocular manifestations of skin conditions: rosacea, herpes zoster
  - Subconjunctival hemorrhage



CLINICAL PRESENTATION AND DIAGNOSIS

- Eye redness
- Eye discharge (watery or pus)
- Itching
- Pain
- Vision problems
- Sensitivity to light (photophobia)
- Swollen eyelids
- Excessive tearing

History	Allergies, contact lens wearers, trauma, discharge (e.g. colour, consistency), pain, photophobia, unilateral or bilateral symptoms, duration of symptoms, previous dermatologic or rheumatologic conditions, recent upper respiratory tract infection
Physical exam	Visual acuity, pupil size, reaction to light, assessment of face/eyelids/conjunctiva/sclera/cornea, intraocular pressure (if concerned with acute glaucoma)

RISK FACTORS AND MEDICAL/DRUG RELATED CAUSES

RISK FACTORS

- Contact lens wearers (extended periods, too tight, not cleaning them)
- Occupation
- Allergen exposure
- Age
- Contact with someone who has a bacterial/viral eye infection (e.g. conjunctivitis)
- Being immunocompromised

MEDICAL RELATED CAUSES

- Conjunctivitis (allergic, bacterial, viral)
- Sjogren's syndrome (reduced tear production)
- Meibomian Gland Dysfunction (MGD)
- Blepharitis
- Iritis
- Acute angle-closure glaucoma
- Herpes simplex, Herpes zoster
- Subconjunctival hemorrhage
- Pterygium (Tissue abnormal overgrowth)

RED FLAGS FOR REFERRAL

- Severe pain
- Vision loss/ significant vision changes/abnormal pupil reaction
- Contact lens wearers
- Corneal involvement
- Trauma to eye
- Recent eye surgery
- Chemical injury
- History of iritis and angle closure glaucoma
- History of chronic conjunctivitis with duration >3 weeks
- Suspected serious infections of COVID-19, Herpes simplex etc.
- Systemic symptoms (e.g. nausea/vomiting, fever, headache)
- No improvement within 48 hours (for bacterial conjunctivitis)

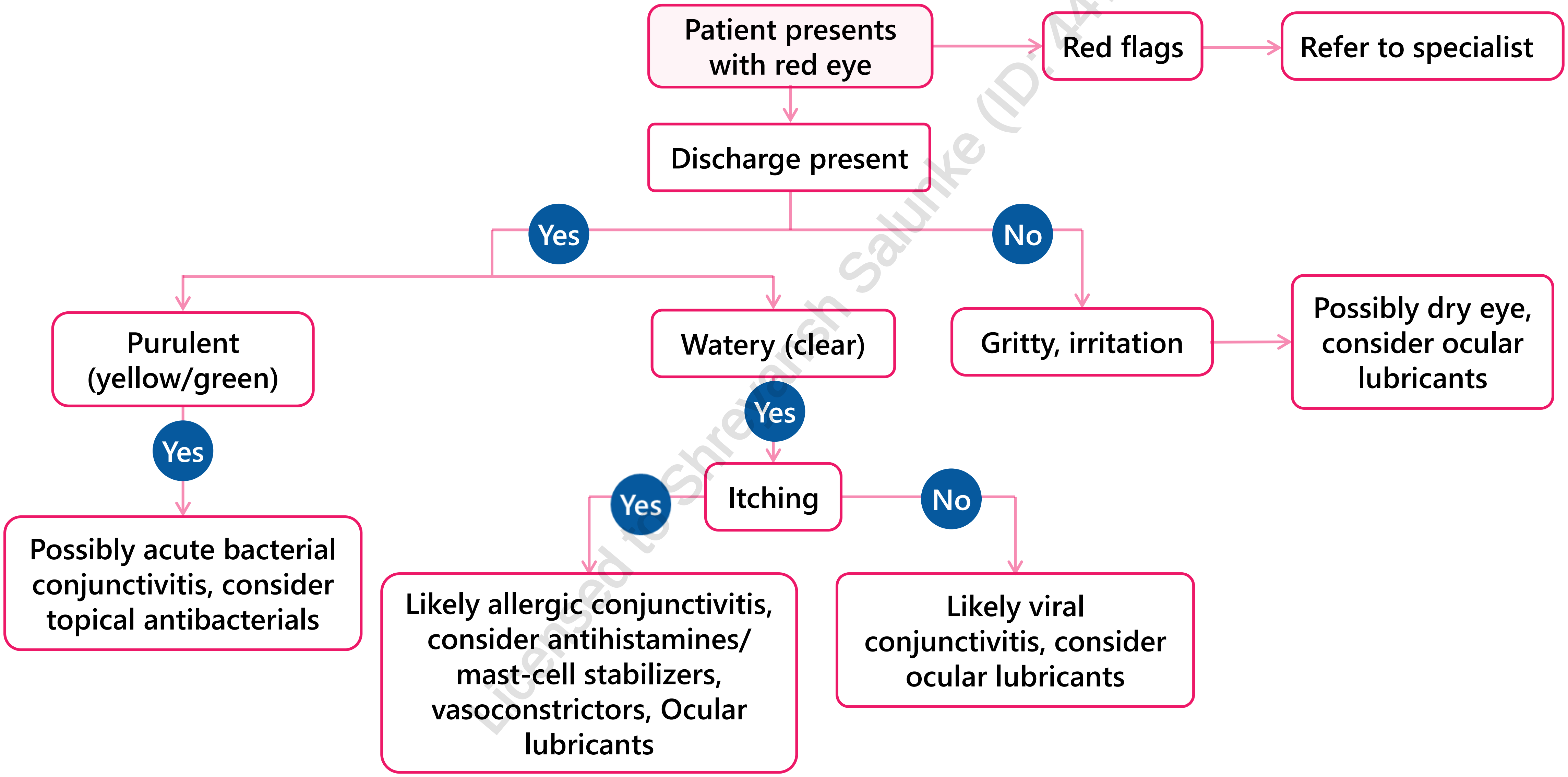
GOALS OF THERAPY

- ✓ Prevent vison loss and structural eye damage
- ✓ Prevent and control spread of eye infection
- ✓ Control swelling
- ✓ Provide relief of present symptoms
- ✓ Treat infection

NON-PHARMACOLOGICAL MEASURES

- Do not wear contact lenses until symptoms/infection has resolved
- Avoid applying eye makeup, discard contaminated eye products
- Avoid contact with irritants (e.g. smoke and wind)
- Patients with bacterial or viral conjunctivitis: Prevent spread by frequent handwashing, using separate towels from other and changing towels and pillowcases after each use
- Patients with bacterial conjunctivitis: apply warm compress 2-3 times daily for 5-10 minutes, followed by gentle massage over lesion to remove crusts and reduce swelling
- Patients with allergic or viral conjunctivitis: apply cool compress to help relieve discomfort
- Patients with allergic conjunctivitis: minimize exposure (stay indoors, keep windows closed, ensure proper ventilation and air filtration systems)

TREATMENT ALGORITHM



BACTERIAL CONJUNCTIVITIS

OVERVIEW

- Cause: Infecting pathogens include *S. aureus* (most common), *S. pneumoniae* (children), H. *influenzae* (children) *M. catarrhalis* (children)
- Symptoms: purulent discharge (thick yellow/white/green) at corners of eye(s), crusty/sticky eyes after sleeping, generalized redness, itchy and burning eyes, foreign body sensation, feeling of fullness around eyes, mild photophobia, usually unilateral
- Spread: through direct eye-to-hand contact
- Treatment:
  - Self-limiting condition but antibiotics can shorten the duration of symptoms if given before day 6
  - Antibiotic treatment may also decrease the risk of more serious corneal complications, recurrences and person-to-person spread
  - Continue agents for at least 2 days after symptom resolution, if no improvement after 48 hours of starting treatment → refer

BACTERIAL CONJUNCTIVITIS

PHARMACOLOGICAL MEASURES

Chronic use of all the agents above may cause corneal epithelial toxicity, allergy and bacterial resistance

Agent	Administration	Comments
Erythromycin ophthalmic ointment 5mg/g	Ointment: 1.25 cm 2–6 times/day x 5-7 days	• Preferred agent in children • May be used in pregnancy
Fusidic acid viscous eyedrops 1% (Fucithalmic®)	1 drop Q12H × 7 days	• Enterobacteriaceae and pseudomonas are resistant to fusidic acid. • Due to its broad-spectrum and BID dosing, fusidic acid is especially useful in children and elderly patients. • Indicated for 2 years of age and older
Gramicidin/polymyxin B (Polysporin® Pink Eye)	Drops: 1-2 drops QID for 4-5 days	• For very mild infections
Tobramycin ophthalmic solution or ointment 0.3% w/v (Tobrex®)	Drops: 1–2 drops Q4H for 7 days Ointment: 1.25 cm BID–TID for 7 days	• For adults and children above 1 years old
Trimethoprim 0.1% & polymyxin B 10,000units/ mL	Drops: 1-2 drops Q3H (maximum 6 times/day) for 7-10 days or as directed by doctor	• Avoid contamination of the dropper

VIRAL CONJUNCTIVITIS

OVERVIEW

- Cause: adenovirus (most common), HSV and HZ
- Symptoms: watery discharge, generalized redness, foreign body sensation, mild photophobia, may have associated respiratory tract infection or shingles
- Spread: highly contagious; respiratory tract-to-eye, finger-to-eye or instrument-to-eye (physician's office) contact, contaminated swimming pools
  - Incubation period ranges from 2–14 days, infection can last from 2–4 weeks
- Treatment: ocular lubricants
  - Severe itching: ophthalmic antihistamines or decongestants
  - Redness, edema: ophthalmic vasoconstrictors (e.g. naphazoline, tetrahydrozoline, phenylephrine) (overuse may cause rebound hyperemia and/or tachyphylaxis), brimonidine 0.025% (brimonidine does not appear to cause significant rebound hyperemia or tachyphylaxis)

ALLERGIC CONJUNCTIVITIS

OVERVIEW

- **Cause**: grass pollen (most common), dry eyes (reduced tear volume leads to decreased capacity to dilute and wash away allergens)
- **Symptoms**: severe bilateral itching, watery discharge, generalized redness, mild eyelid swelling
- **Spread**: not contagious
- **Treatment**: ocular lubricants, saline, ophthalmic decongestants, mast cell stabilizers, ophthalmic antihistamines

OCULAR LUBRICANTS

FYI

PHARMACOLOGICAL MEASURES

- All ocular lubricants are composed of an aqueous base but differ in other ingredients and properties
- Examples of ocular lubricants:
  Clinical trials do not provide definitive conclusions about which lubricant is best – use a trial-and-error approach
    - Carboxymethylcellulose
    - Dextran 70/hypromellose (hydroxypropyl methylcellulose)
    - Hypromellose (hydroxypropyl methylcellulose)
    - Mineral oil/petrolatum
    - Polyvinyl alcohol
    - Propylene glycol/polyethylene glycol-400
    - Sodium hyaluronate
- Preservatives can cause adverse effects → consider preservative-free lubricants for patients with irritated eyes who require drops > 4-5/daily
    - Examples of preservatives: sodium chlorite, sodium perborate, cetrimide, polyquaternium-1, benzalkonium chloride (limit use to 4 applications/day)
    - Preservative free eye drops: Hydrasense®, Hylo®, Refresh® (celluvisc)
- Gels can cause blurry vision → counsel patients to use at bedtime

OCULAR LUBRICANTS

FYI

PHARMACOLOGICAL MEASURES

Property of Ocular Lubricants	Comments	Examples
Viscosity	Although several viscosity agents are available, there is no evidence to inform if one is more effective than the other Higher viscosity agents (e.g., petrolatum ointments, carboxymethylcellulose 1%) increase tear retention time on the ocular surface but may cause more visual disturbance than lower viscosity agents.	Carbomer 940 (polyacrylic acid) Carboxymethylcellulose Dextran, Hyaluronic acid Hydroxypropyl-guar (HP-guar) Hydroxypropyl methylcellulose Petrolatum, Polyethylene glycol Polyvinyl alcohol, Polyvinyl pyrrolidone (povidone)
Lipids	To replace low lipid levels and prevent tear evaporation	Dimyristoylphosphatidylglycerol (DMPG) Castor oil, Mineral oil
Preservatives	Benzalkonium chloride, known to be toxic to the epithelium. Limit use to 4 applications/day or use the following alternatives: • Lubricants with other less toxic preservatives (see examples) • Unit dose lubricants • Multi-dose bottles with filter/valve to maintain sterility (no preservatives included) • Ointments, which do not support bacterial growth and do not require preservatives. However, these can produce unacceptable blurred vision	Cetrimide Polyquaternium-1 (Polyquad) Sodium chlorite (Purite) Sodium perborate (GenAqua)
Electrolytes	Adding electrolytes to ocular lubricants may better mimic natural tears. In natural tears, potassium helps to maintain corneal thickness while bicarbonate aids in the recovery of epithelial barrier function.	Calcium chloride, Magnesium chloride, Potassium chloride, Sodium bicarbonate, Sodium chloride

VIRAL AND ALLERGIC CONJUNCTIVITIS TOPICAL PHARMACOLOGICAL THERAPY

Agent	Administration	Comments
Mast cell stabilizers e.g. sodium cromoglycate	Must start before allergy season	Symptom relief takes 2-3 days (start before allergy season). May use in pregnancy.
Antihistamines e.g. olopatadine, ketotifen, bepotastine	Patients may experience minor stinging upon instillation	Work faster than oral antihistamines, fewer adverse effects. May use in pregnancy.
Antihistamine and decongestant combination	Do not use for more than 3 consecutive days as this can cause rebound symptoms. Patients may experience minor stinging upon instillation.	Second-line. Use is generally not recommended during pregnancy. Decongestants CI in narrow angle glaucoma.
Topical decongestants e.g. naphazoline, tetrahydrozoline	Do not use for more than 3 consecutive days as this can cause rebound symptoms	

MONITORING PARAMETERS

Eye condition	Reassessment
Bacterial Conjunctivitis	Adults: Refer if worsening or no improvement within 48 hours Children: requires consultation for further therapy. Do not recommend self-treatment.
Viral Conjunctivitis (Adenovirus)	Requires medical assessment to rule out other viral causes. If decongestant or lubricant drops cause irritation, discontinue use or suggest a preservative-free product.
Seasonal Allergic Conjunctivitis	If decongestant or lubricant drops cause irritation, discontinue use or suggest a preservative-free product. Refer if severe symptoms or no improvement within 3 days or if symptoms persist despite treatment.

COUNSELLING

Suspension eye drops should be shaken well before use

Remove contact lenses prior to instilling drops & wait at least 15 min after instilling drops to insert contact lens

If the applicator of the drop touches a surface, wipe with a clean tissue. Do not rinse tip of bottle with water as this will change the solution's stability and composition.

1. Wash hands and pull down lower eyelid using thumb and index finger to form a pouch
2. Instill only 1 drop and close eye without squeezing or blinking for 30-60 seconds to allow absorption—use nasolacrimal duct occlusion by applying gentle pressure with finger across the bridge of the nose to prevent systemic absorption
3. Wait 3-5 min between administration of different eyedrops to prevent flushing away or diluting the first drop—if you are using both eye drops and eye ointment, apply the eyedrop at least 10 minutes before the ointment.
4. Remove excess fluid gently with a clean tissue—do not rub
5. Discard bottle after 28 days after opening

Refer to the product monograph for specific instructions on proper use, as information may vary.

CASE

FJ is a 21-year-old male who presents to your family health team with an eye infection. FJ has yellow discharge in his left eye and had a hard time opening it this morning due to crust formation. He states it is also a little irritated, but he isn't in pain. The itchiness started yesterday.

1. All of the following are appropriate non-pharmacological recommendations EXCEPT:

   a) Eyelid hygiene prior to bed
   b) Apply a cool compress
   c) Apply a warm compress
   d) Avoid contact with irritants

CASE

FJ is a 21-year-old male who presents to your family health team with an eye infection. FJ has yellow discharge in his left eye and had a hard time opening it this morning due to crust formation. He states it is also a little irritated, but he isn't in pain. The itchiness started yesterday.

2. What is the most appropriate pharmacologic recommendation for FJ?

   a) Erythromycin ointment in the affected eye
   b) Tetrahydrozoline eye drops in the affected eye
   c) Ketorolac eye drops in the affected eye
   d) Olopatadine eye drops in the affected eye

CASE

FJ is a 21-year-old male who presents to your family health team with an eye infection. FJ has yellow discharge in his left eye and had a hard time opening it this morning due to crust formation. He states it is also a little irritated, but he isn't in pain. The itchiness started yesterday.

3. Which of the following is correct regarding eyedrops?
   a) Wait 3-5 minutes between drops of the same medication
   b) Wait 1-2 minutes between drops of different medications
   c) Discard after 4 weeks
   d) Clean tip of bottle with water

CASE

FJ is a 21-year-old male who presents to your family health team with an eye infection. FJ has yellow discharge in his left eye and had a hard time opening it this morning due to crust formation. He states it is also a little irritated, but he isn't in pain. The itchiness started yesterday.

4. When can FJ see some benefit from the antibiotics?
   a) Within 10 days
   b) Within 7 days
   c) Within 5 days
   d) Within 3 days

EYELID CONDITIONS

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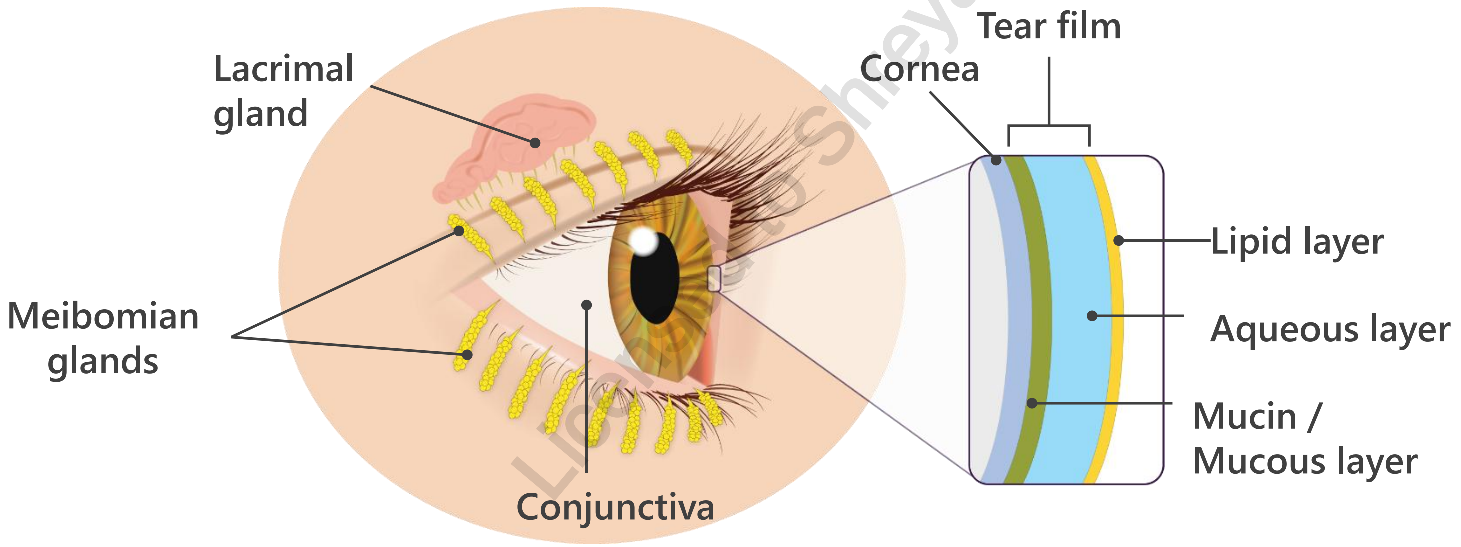
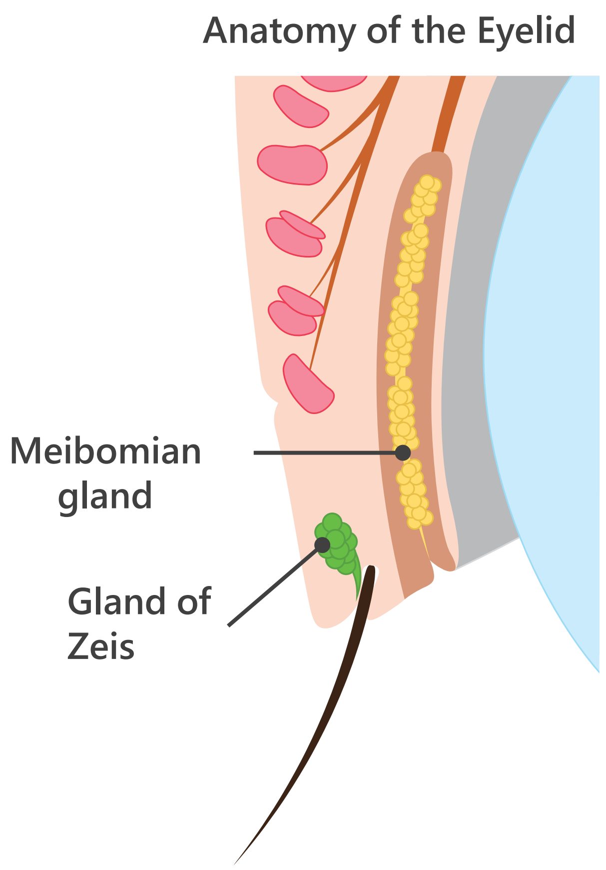
PATHOPHYSIOLOGY

- Eyelids and lashes protect the eyeball from foreign bodies and injuries and help maintain a wet corneal surface. Underneath the eyelid skin lies loose tissue that is capable of significant edema and swelling.
- Meibomian glands: sebaceous gland that secretes lipid layer of the tear film to keep tears from drying too quickly
- Gland of Zeis: sebaceous glands that produce oily substance that is associated with lash follicles to keep them from becoming brittle
- Gland of Moll: sweat glands



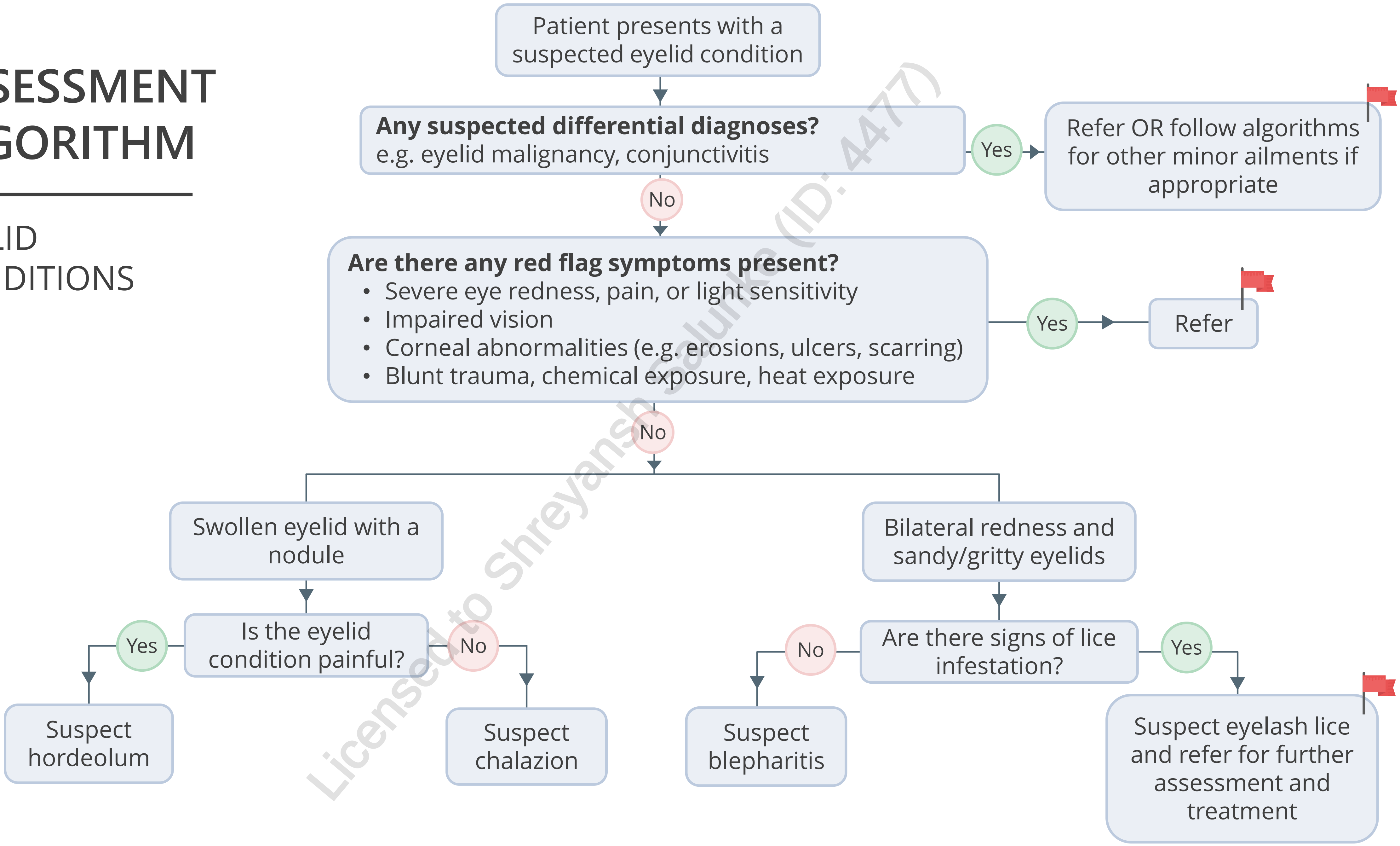


GOALS OF THERAPY

- ✓ Resolve infection
- ✓ Prevent recurrence
- ✓ Prevent transmission to other eye or to household contacts
- ✓ Reduce inflammation and discomfort
- ✓ Reduce risk of complications

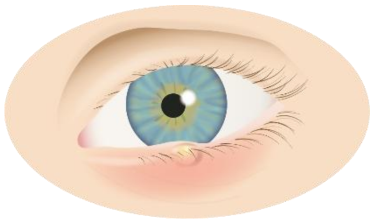
ASSESSMENT ALGORITHM

EYELID CONDITIONS

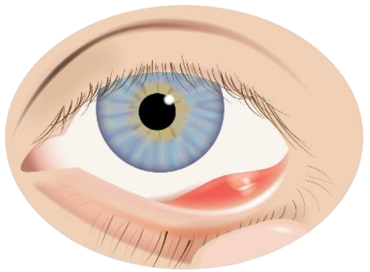


HORDEOLUM (STYE)

- Cause: *Staphylococcus aureus* infection of the sebaceous glands of the eyelid
- Symptoms: unilateral, localized lid swelling, painful, tender
  - External Stye: sebaceous glands of Zeis (associated with lash follicle), infection is smaller, more superficial, lesion points towards skin
  - Internal stye: meibomian sebaceous glands, infection is larger and more swollen, points either to skin or to conjunctival surface, more prolonged course compared to external
- Treatment:
  - Usually resolves spontaneously (external- 48 hours, internal- 1-2 weeks)
  - Medication is usually not necessary and is not recommended
  - Warm compresses for 10-15 minutes 3-4 times daily can hasten resolution—massage the eyelid toward the lid margin
  - Once the stye drains, remove the excess discharge by cleaning the eyelid with warm water and a face towel, or eyelid wipes
  - Referral: Incision/drainage may be required if no improvement or if stye has not spontaneously drained within usual time. May need to use anti-bacterial ointment afterwards to prevent further infection
  - See non-pharmacological measures in Red Eye section for information on preventing transmission



External hordeolum



Internal hordeolum

CHALAZION

OVERVIEW & PHARMACOLOGICAL MEASURES

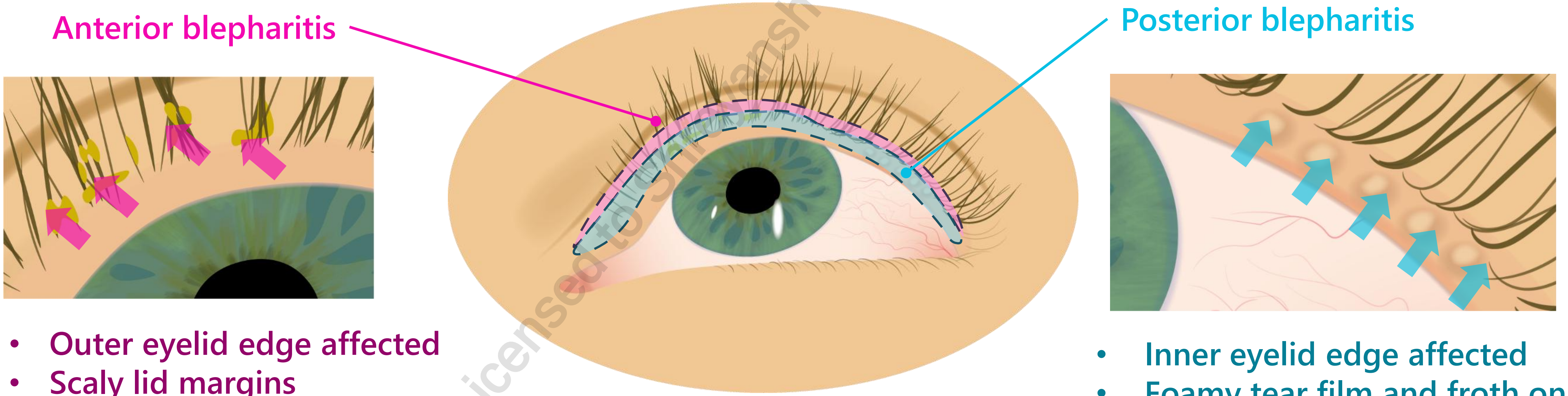
- Cause: sterile, idiopathic chronic meibomian gland inflammation; blockage of meibomian gland orifices causes stagnation of sebaceous gland secretions
  Symptoms: painless (distinguishing it from styes), localized eyelid swelling, red lesion/nodule, unilateral, conjunctival irritation, larger chalazia may press on eyeball and cause astigmatism or visual distortions
- Treatment
  - Non-pharmacological: warm compress to soften sebaceous gland secretions that are blocking meibomian gland orifices—massage eyelid toward lid margin. Once chalazion drains, remove discharge
  - Referral: Immediate referral for eye pain or impaired vision; refer if lesion not resolved in few days
  - Pharmacological: not recommended; larger chalazia may require surgical excision, intralesional steroid injections or both

Chalazion

BLEPHARITIS

Cause: altered lipid composition in meibomian gland secretions causes unstable tear film. This promotes bacterial growth (*S. aureus* and *S. epidermidis*). May also be caused by contact dermatitis.

- Symptoms: bilateral irritation, burning, itching, swelling, scaling of lid margins, sandy/gritty sensation that is worse upon awakening.



- Outer eyelid edge affected
- Scaly lid margins
- Bacterial and/or seborrheic

- Inner eyelid edge affected
- Foamy tear film and froth on lid margin

BLEPHARITIS

Treatment:

- Regular and long-term eyelid margin hygiene: 2–3 times daily immediately after initial diagnosis or during periods of exacerbation, but may be reduced to twice a week once control has been achieved
- Warm compresses: apply to closed eyelids for 5–10 minutes to melt solidified material in the glands
- Artificial tears: may help to alleviate symptoms, especially in patients with associated dry eye disease
- May require topical or oral antibiotics, and in refractory cases, topical glucocorticoids or topical cyclosporine
- Avoid any commercial products that contain cocamidopropyl betaine (CAPB) as this can cause contact dermatitis, and eyelid dermatitis

PHARMACOLOGICAL TREATMENT OF BLEPHARITIS

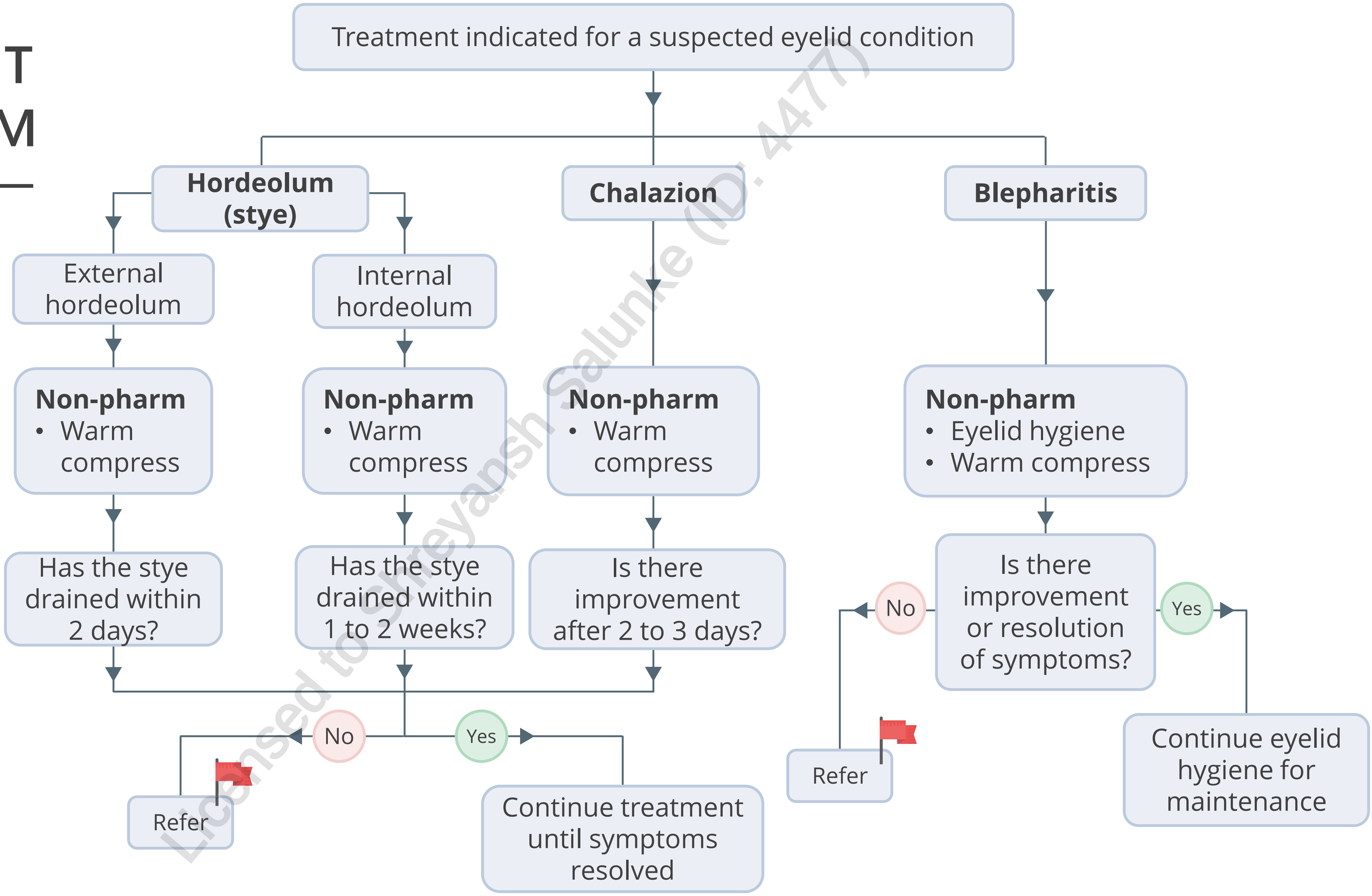
MEDICATION	ADMINISTRATION	ADVERSE EFFECTS AND CONSIDERATIONS
Erythromycin (topical ophthalmic antibiotic ointment)	Apply directly onto the lid margin 1 to 4 times a day for 1 to 2 weeks, once symptoms improve reduce to daily at bedtime for 4 to 8 weeks	• Generally well-tolerated • Broad spectrum antimicrobial activity • More effective in anterior blepharitis
Doxycycline Minocycline Tetracycline (oral antibiotics)	• Doxycycline: 100 mg daily • Minocycline: 50 to 100 mg daily • Tetracycline: 250 mg QID	• Reserved for chronic moderate to severe blepharitis that have inadequate response to topical antibiotic therapy • Contraindicated in pregnancy, nursing women and children <8-13 years of age depending on the agent • GI effects, yeast overgrowth, photosensitivity • Products containing Al, Ca, Mg, or Fe may impair the absorption of oral tetracyclines (separate doses by 2 hours)
Azithromycin Erythromycin (oral macrolide antibiotics)	• Azithromycin: 1 g weekly or 500 mg daily for 3 days, repeated weekly • Erythromycin: 250 mg QID	• GI effects (nausea/vomiting, diarrhea), QT_c interval prolongation • Alternative to tetracycline if patient is pregnant, nursing or <8-13 years of age

PHARMACOLOGICAL TREATMENT OF BLEPHARITIS

MEDICATION	ADMINISTRATION	ADVERSE EFFECTS AND CONSIDERATIONS
Loteprednol Fluorometholone (topical ophthalmic low-potency glucocorticoids)	• Limit use to 2 to 3 weeks	• Reserved for patients with blepharitis that is unresponsive to other therapies • Risk for increased intraocular pressures, cataract formation or glaucoma
Topical Cyclosporine	• 0.05% eye drops, "off-label" for blepharitis in the absence of dry eyes	• Reserved for patients with blepharitis that is unresponsive to other therapies, alternative to topical steroids • Eye pain (1% to 22%), burning sensation of eyes (17%)

TREATMENT ALGORITHM

EYELID CONDITIONS



MONITORING PARAMETERS

Eye lid condition	Goal and endpoint of therapy
Stye	• Spontaneous drainage within 48 h
Chalazion	• Improvement should begin within a few days. Complete resolution can take weeks to months.
Blepharitis	• Control inflammation and discomfort • Reduce risk of severe, long-term complications

SUMMARY OF DIFFERENCES AMONG EYE LID CONDITIONS

Eye lid condition	Description	Onset	Signs and symptoms
Stye	Acute infection of 1 or more eyelid glands (meibomian, Zeis)	Acute (days)	Unilateral painful lesion, localized lid swelling, tenderness, erythema
Chalazion	Sterile, focal, chronic inflammation of the lid due to obstructed meibomian gland	Gradual enlargement (days to weeks)	Non-tender (painless), rubbery nodule, localized lid swelling, often unilateral
Blepharitis	Chronic inflammation of the lid margins associated with infection, dermatological conditions or meibomian gland dysfunction	Gradual irritation (weeks to months)	Irritated, possibly reddened lid margins; greasy, scaly and/or flaky; usually bilateral

CASE

Simon comes to your pharmacy complaining of his irritated eyes. He tells you that both his eyes are itchy and feel most gritty in the morning. You also notice dandruff-like material on his eyelids.

1. Which condition is Simon likely presenting with?
   a) Dry eyes
   b) Blepharitis
   c) Hordeolum
   d) Viral Conjunctivitis

CASE

Simon comes to your pharmacy complaining of his irritated eyes. He tell you both his eyes are itchy and feel most gritty in the morning. You also notice dandruff-like material on his eyelids.

2. What would be the MOST effective pharmacological treatment?
   a) Antibacterial eye ointment
   b) Antibacterial eye drops
   c) Antihistamine eye drops
   d) Lubricating eye drops

CLINICAL PEARLS AND TAKEAWAYS

- Some patients may react to preservatives in ophthalmic products; lubricants without preservatives are available as single-dose preservative-free containers or in multidose bottles
    - These formulations are preferred for patients who require drops more than 4–5 times daily for dry eye disease, those with allergic conjunctivitis or those who experience eye irritation with preserved medication
- Use of ophthalmic corticosteroids may cause glaucoma and/or cataracts; the risk increases with prolonged use
- Ophthalmic corticosteroids and antibiotic/corticosteroid combinations may worsen herpetic/fungal keratitis and should be used only under the supervision of an ophthalmologist
    - Non-steroidal anti-inflammatory drugs (NSAIDs) can cause keratitis, corneal melting and perforation in patients who have pre-existing corneal disease and, in these cases, should be used under the supervision of an ophthalmologist.
- Topical decongestants/vasoconstrictors may provoke angle-closure glaucoma in patients who are predisposed (see glaucoma lecture)
- Topical anesthetic eye drops should never be prescribed for self-administration by the patient

AGE-RELATED MACULAR DEGENERATION

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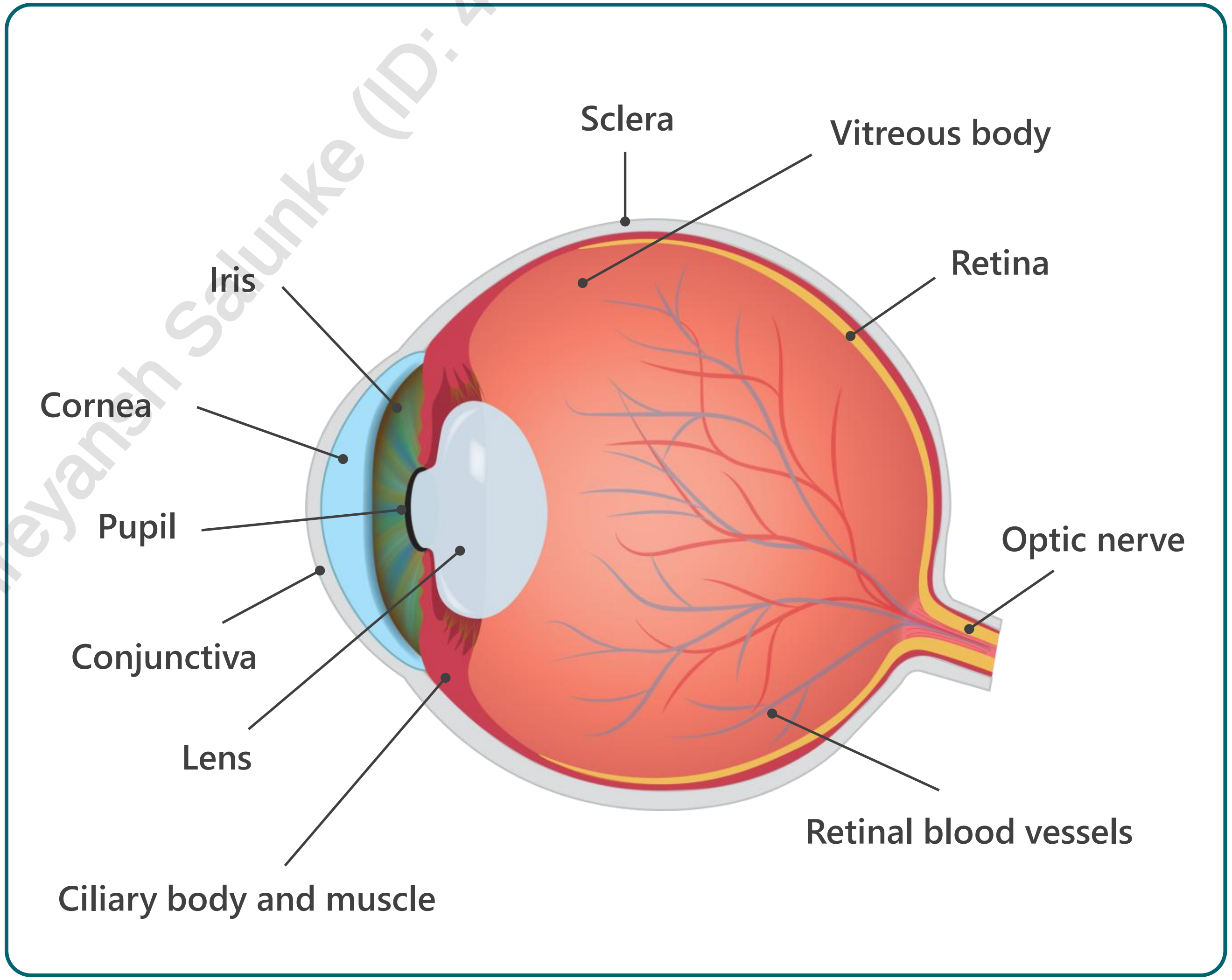
PATHOPHYSIOLOGY

- Age-related macular degeneration (AMD) is a common eye condition characterized by damage/deterioration of the macula (located in the center of the retina and required for sharp, central vision)
- AMD can be classified into two categories:
    - Dry (nonexudative)
    - Wet (Neovascular AMD, exudative): responsible for severe vision loss if not treated in a timely manner



CLASSIFICATION

	Dry	Wet
Prevalence	85-90% of cases	10-15% of cases
Severity	Less severe, more associated with aging	Severe, can lead to vision loss
Progression	Slowly progresses (years), can unpredictably change to wet AMD	Rapid
Features & pathophysiology	• Have pigmented layer of cells next to retina called Retinal Pigment Epithelium (RPE). Thinning of this layer causes AMD. • Clumping under RPE causes drusen deposits • Drusen are white/yellow spots in retina caused by protein and lipids accumulating • Drusen can be hard or soft • Drusen do not cause AMD but having them increases chances of developing it	• Development of Choroidal Neovascular Membrane (CNV) • This is growth of abnormal blood vessels beneath and sometimes into the retina, usually from choroidal layer • Accumulation of fluid/blood beneath retina • Sudden bleeding/leaking from weak blood vessels, creating a scar leading to permanent vision loss

CLINCAL PRESENTATION

- Early AMD may be asymptomatic
- Painless
- Distortion of straight lines
- Blurred vision
- Loss of central vision in one or both eyes
- Dark patch in central vision (Scotoma)
- Impacts quality of life
    - Driving
    - Reading
    - Daily activities
- Depression

Age-related Macular Degeneration

RISK FACTORS

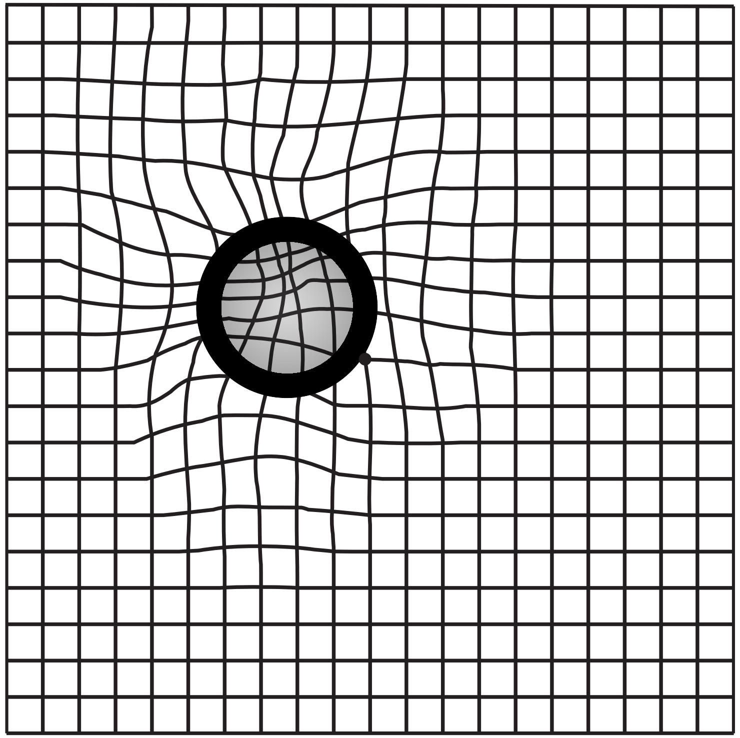
- Aging (especially > 65 years old)
- Smoking
- Caucasians
- Family History and genetics
- Hypertension
- Cardiovascular disease

The Canadian Ophthalmological Society recommends that all people ages 41 to 55 have a comprehensive eye examination at least every 5 years to detect AMD early

DIAGNOSIS

- Medical History
  - Recent changes in vision
- Physical examination
  - Visual acuity test
  - Amsler grid
  - Funduscopy (biomicroscopy)
  - Fluorescein angiography (image)
    - Dye injected and retinal photographs taken
  - Optical coherence tomography- non-invasive cross-sectional images of retina

Amsler grid



GOALS OF THERAPY

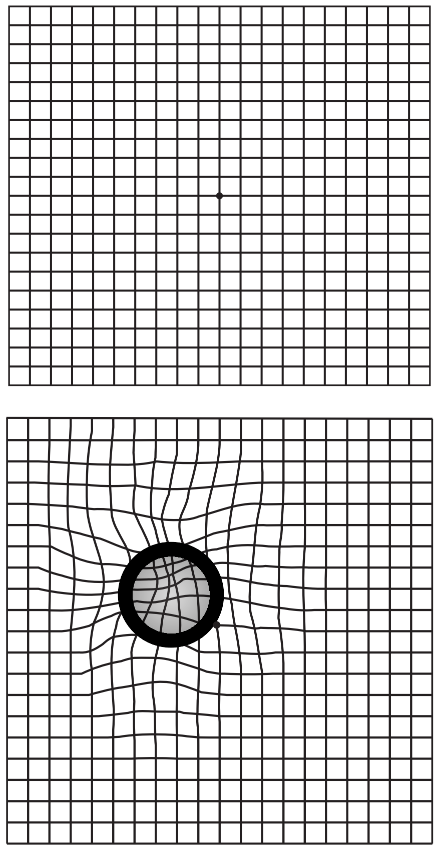
- ✓ Minimize vision loss
- ✓ Improve vision
- ✓ Decrease central scotoma enlargement (Central blur in the center of visual field)
- ✓ Enhance functional capabilities
- ✓ Minimize impact on quality of life

NON-PHARMACOLOGICAL MEASURES

- Smoking cessation – smoking is a major risk factor for AMD
- Regulate blood pressure
- Regulate cholesterol levels
- Exercise
- Balanced diet

The Amsler grid is useful for self-monitoring by patients:

1. Make sure the room is well lit
2. Hold the grid at a normal reading distance (30 - 35 cm) away from the face
3. Wear reading glasses if typically required
4. Cover one eye
5. Focus on the dot in the middle
6. Check to see if any of the lines on the grid are missing, blurry, curved or broken; or if the borders of the grid are not visible in any part



PREVENTATIVE THERAPIES

- **Age-Related Eye Disease Study (AREDS) Formulation**
  - Decrease risk of advanced AMD and related loss of vision by 25% in those with intermediate dry AMD
  - Does not restore vision and ineffective in early AMD
  - Does not prevent onset of AMD
  - *β- carotene is no longer recommended due to a small increase in risk of lung cancer seen in patients who took formulations containing beta carotene → this risk was observed at a higher rate in smokers and former smokers

- AREDS 2 trialled different formulations showing no significant difference but demonstrated removing β- carotene had similar efficacy and a lower zinc dose was as effective and better tolerated → preference to use the AREDS 2 formulations

Many products containing lutein, zeaxanthin and omega 3 are now marketed for improving eye health; omega supplements were shown to be of no value in AMD

	Standard Formulation AREDS	AREDS 2 (more recent study)
Other Ingredients	Vitamin C 500mg, Vitamin E 400U, Zinc 80mg, Copper 2mg, β- carotene* 15mg	• Addition of Lutein + Zeaxanthin, omega 3 + omega 6, or all 4 to the original AREDS formulation (No B-carotene) • No additional reduction in vision loss

PHARMACOLOGICAL MEASURES

WET AMD

Vascular Endothelial Growth Factor (VEGF) inhibitors

Bevacizumab (Avastin®)	• Recombinant humanized monoclonal antibody • Commonly used off-label for AMD (primary use colon cancer) • Less expensive than ranibizumab with no therapeutic difference
Ranibizumab (Lucentis®)	• Antibody fragment of bevacizumab; excellent evidence
Aflibercept (Eylea®)	• Recombinant fusion protein; no significant therapeutic difference compared to ranibizumab (no direct trials against bevacizumab), widely used as second line

Decrease vision loss by inhibiting growth and shrinking new blood vessels, reduce edema. High recurrence rate

- Administration: injected directly into the vitreous cavity of the eye every 4 weeks
  - Use topical anaesthetic and antiseptic (povidone-iodine or chlorhexidine if allergic to povidone-iodine)
- Safety: side effects are uncommon aside from mild eye pain and redness
  - Sub-conjunctival hemorrhage may occur and resolve spontaneously in 2 weeks
  - Bacterial Endophthalmitis (severe vision loss) is rare but is a medical emergency

PHARMACOLOGICAL MEASURES

WET AMD

Photodynamic Therapy With Verteporfin

Verteporfin is administered as IV injection followed by administration of laser light to the affected eye

- Dye is activated by laser light and oxygen-forming free radicals
- Impairs neovascular endothelium and prevents vessels from leaking

- Efficacy: no longer used as primary therapy for most AMD forms
  - Does not restore or prevent vision loss
  - Less effective and more complicated to administer compared with the VEGF inhibitors
  - Used in rare forms of neovascular AMD
- Safety: must avoid bright light for 48 hours after treatment due to light sensitivity → can lead to severe burns if exposed to light due to verteporfin.
  - Back pain can occur during verteporfin infusion. Temporary visual disturbances can occur following therapy
- Drug interactions: increased risk of photosensitive with other photosensitizing medications such as tetracyclines

CLINICAL PEARLS AND SUMMARY

- Dry AMD
  - No proven efficacious drug therapy for dry AMD
  - Smoking cessation
  - Try AREDS formulation or AREDS 2 formulation (Vitalux® Advanced) to prevent progression from intermediate AMD to advanced AMD. (Obtain smoking history first). If patient is taking other vitamins or minerals, ensure the patient does not exceed the total daily amount recommended. Take with food to reduce nausea that may be associated with zinc.
  - Caution: Orlistat can reduce the absorption of fat-soluble vitamins and beta-carotene. Should separate administration by 2 hours

- Wet AMD
  - VEGF inhibitors

Refer to physician for any vision loss or change

CASE

LP is a 65-year-old female (BMI= 30 kg/m2) who was just diagnosed with intermediate dry AMD. She comes to your pharmacy and asks the pharmacist to explain what AMD is and answer some questions she has about it. Her past medical history is significant for dyslipidemia and hypertension. She currently takes ramipril 10 mg PO once daily, amlodipine 5 mg PO once daily, and atorvastatin 40 mg PO once daily. She walks for 30 minutes once per week and does not smoke. She has no allergies to medications. Today, her blood pressure is 165/90 mmHg and her LDL-C is 2.3 mmol/L.

1. All of the following are appropriate non-pharmacological recommendations for LP EXCEPT:

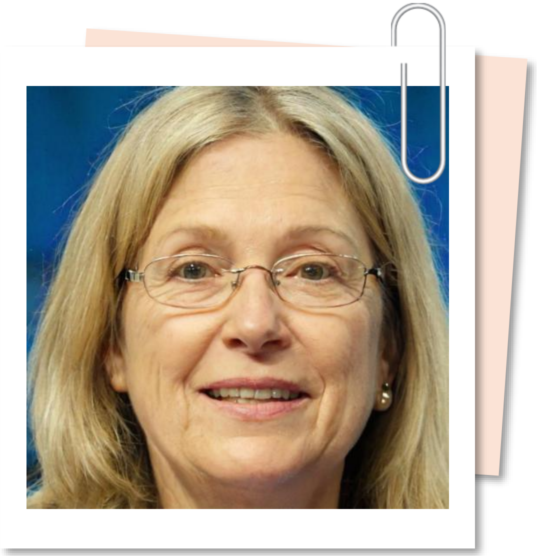
   a) Blood pressure control
   b) LDL reduction
   c) Smoking cessation
   d) Increase physical activity

CASE

LP is a 65-year-old female (BMI= 30 kg/m2) who was just diagnosed with intermediate dry AMD. She comes to your pharmacy and asks the pharmacist to explain AMD and answer some questions she had about AMD. Her past medical history is significant for dyslipidemia and hypertension. She currently takes ramipril 10 mg PO once daily, amlodipine 5 mg PO once daily, and atorvastatin 40 mg PO once daily. She walks for 30 minutes once per week and does not smoke. She has no allergies to medications. Today, her blood pressure is 165/90 mmHg and her LDL-C is 2.3 mmol/L.

2. What is the MOST appropriate recommendation for LP?

   a) Vitamin C, Vitamin E, Zinc, Copper, β- carotene*
   b) Vitamin C, Vitamin E, Zinc, Copper, Zeaxanthin, Omega 6
   c) Vitamin C, Vitamin E, Zinc, Copper, Lutein, Omega 3
   d) Vitamin C, Vitamin E, Zinc, Copper, Lutein, Zeaxanthin, Omega 3 and 6



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3. Which of the following is false?
   a) LP's new supplements should reduce the risk of advanced AMD
   b) LP's new supplements should restore her vision
   c) AREDS2 utilizes a lower zinc dose to improve tolerability compared to the standard AREDS formulation
   d) Removing β- carotene from the standard AREDS formulation showed similar efficacy

CASE

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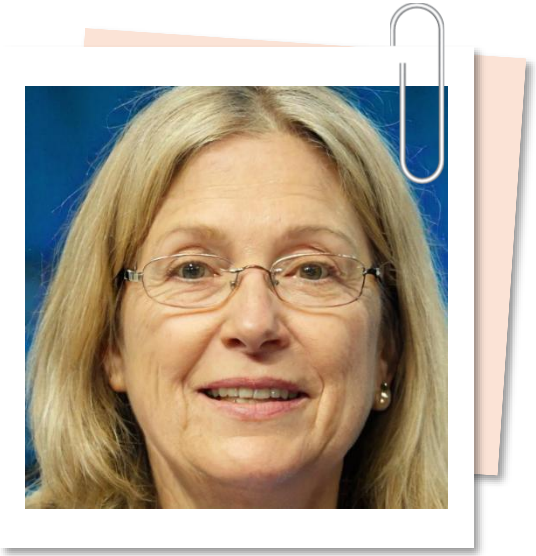
4. If LP's right eye progressed to wet AMD, which of the following would be most appropriate?

   a) Bevacizumab
   b) Etanercept
   c) Pembrolizumab
   d) Rituximab



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CHANGE LOG

March 2025
- Slide 3: Updated legend
- Slide 19: Removed ophthalmic oxymetazoline as discontinued
- Slide 21: Added disclaimer at the bottom that instructions may vary
- Slide 34: Removed bacitracin as a treatment option for blepharitis as single-ingredient product is discontinued

April 2025
- Slide 34: Updated age restrictions for tetracyclines

May 2025
- Slide 19: Changed pregnancy recommendation

August 2025
- Slide 14: Added strength of trimethoprim/polymyxin and minor edit to dosage
- Slide 16: Added ophthalmic antihistamines
- Slide 33: Removed point on omega-3 since mixed evidence
- Minor formatting changes made throughout

November 2025
- Minor formatting edits, no content changes made

February 2026
- Minor formatting changes made
- Slide 2: Updated NAPRA competencies
- Slide 9: Updated red flags
- Slide 19: Added example of mast cell stabilizer (sodium cromoglycate)